Balu S Unny

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PROFILE

Data Analytics postgraduate in Computer Science with hands-on experience in data analysis, modeling, and dashboarding on real-world e-commerce datasets of 540K+ transactions. Built end-to-end projects covering data cleaning, customer segmentation, classification models, and Power BI reporting. Skilled in Python, SQL, Power BI, and Machine Learning, with a focus on turning data into clear, actionable insights.

SKILLS

Languages: Python, SQL
Databases: MS SQL Server, MySQL
Data Analysis: Pandas, NumPy, EDA, Data Cleaning, Statistical Analysis, Feature Engineering
Machine Learning: Scikit-learn, Classification, Regression, Predictive Modeling, Model Evaluation
BI & Visualization: Power BI, DAX, Power Query, Data Modeling, Tableau, Matplotlib, Seaborn, Plotly
Tools: Excel, Git, GitHub, Jupyter Notebook

PROJECTS

E-Commerce Customer Intelligence | Python, RFM Analysis, Machine Learning, Market Basket Analysis

- Analyzed 541K+ e-commerce transactions using EDA, customer segmentation (RFM), and correlation analysis to identify customer behaviour patterns, revealing 47.9% of customers were Hibernating or Lost.
- Built and evaluated classification models on imbalanced customer data using SMOTE and Random Forest, achieving 73% recall and selecting the model based on performance evaluation.
- Applied Apriori Market Basket Analysis across 18,480 baskets and 3,876 products, generating 76 association rules for cross-selling opportunities.
- Identified first-purchase behaviour as the strongest value predictor, accounting for 86% of feature importance.

Sales Performance Analytics | SQL, Power BI, Power Query, DAX, Data Modeling

- Cleaned, validated, and transformed 100K transactional records across 9 relational datasets, ensuring data quality for analysis and integrating data for downstream modeling and reporting.
- Built a Power BI data model and DAX KPI framework tracking sales, order fulfillment, and operational performance across R$13.6M sales, 99.4K orders, and 96K customers.
- Identified a 3.1% repeat purchase rate across 96K customers, highlighting a significant customer retention opportunity.
- Analyzed delivery performance and identified a customer rating decline from 4.3 to 2.6 for on-time versus late deliveries, quantifying the impact of delivery reliability.

EDUCATION

Master of Science in Computer Science (Data Analytics) 2022-2024
Rajagiri College of Social Sciences Kalamassery, Kerala

Bachelor of Technology in Computer Science 2016-2021
College of Engineering and Management Punnapra, Kerala

CERTIFICATIONS

Microsoft Power BI: The Practical Guide, Udemy August 2023
Data Analytics Foundations, Coursera June 2023
Machine Learning Foundations: A Case Study Approach, Coursera May 2023